

Athiyaman M

athiyamanpro@gmail.com | +91-8940089643 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Software Engineer with 1+ year of experience building backend systems, healthcare software applications, AI-driven solutions, and database-intensive platforms. Currently working at C-DAC on national-scale software infrastructure projects involving Python, PostgreSQL, REST APIs, analytics pipelines, log processing, and backend automation. Previously contributed to healthcare technology development at HTIC, IIT Madras, building clinical software using Qt, VTK, and ITK.

Experienced in Python development, SQL optimization, AI/LLM applications, Elasticsearch, ETL workflows, and healthcare data systems. Passionate about building AI-driven healthcare systems, clinical decision support tools and intelligent backend platforms for MedTech applications.

Experience

Project Engineer - Software Engineering

08/2025 – Present | Chennai, India.

Centre for Development of Advanced Computing (CDAC), Government of India

- Developed Python-based backend modules for SSC examination infrastructure supporting security flag detection, score computation, candidate validation, and automated reporting workflows. Built REST API integrations for data exchange, analytics processing, and backend automation services.
- Developed Elasticsearch log parsing pipelines for large-scale production systems, enabling structured monitoring, anomaly detection, and reporting. Automated operational workflows using Python scripts, reducing manual validation effort and improving system reliability.
- Contributed to BharatDB initiatives involving PostgreSQL ecosystem enhancement, database research, and open-source database technologies. Designed and optimized PostgreSQL database operations including query development, data analysis, validation processes, and performance tuning.
- Participated in AI/LLM proof-of-concepts including document intelligence, research summarization, and healthcare-focused AI applications.

Project Intern - Healthcare Software Development

06/2024 – 07/2025 | Chennai, India.

Healthcare Technology Innovation Centre (HTIC), IIT Madras

- Developed healthcare desktop applications for Total Knee Arthroplasty (TKA) using Qt, C++, VTK, and ITK frameworks.
- Worked on medical visualization and image-processing workflows for orthopedic clinical applications.
- Integrated MySQL databases and REST APIs for healthcare data management and application workflows.
- Utilized Azure DevOps, CMake, and CI/CD pipelines for software testing, version control, and deployment processes.
- Collaborated with researchers and engineers in a healthcare innovation environment focused on clinical software solutions.

Projects

[Mental Health Risk Prediction using Explainable AI \(XAI\)](#) | [GitHub](#)

- Developed NLP-based mental health risk prediction system using transformer embeddings and machine learning models.
- Achieved 87% AUC-ROC while benchmarking multiple classifiers.
- Implemented Explainable AI techniques including SHAP, LIME, and Counterfactual Explanations.
- Designed Streamlit dashboard for prediction visualization and interpretability.
- Stack : Python, BERT, Scikit-Learn, SHAP, LIME, Pandas, Streamlit

[Medical AI Assistant using RAG and Llama-2](#) | [GitHub](#)

- Built an AI-powered medical knowledge assistant using Retrieval-Augmented Generation (RAG) architecture.
- Implemented document ingestion, chunking, embedding generation, vector search, and context-aware response generation.
- Developed semantic retrieval workflows using LangChain and Pinecone Vector Database.
- Built Flask-based web application enabling natural-language medical question answering.
- Stack : Python, LangChain, Llama-2, Pinecone, Flask, RAG, NLP.

[Agentic AI Research Assistant](#) | [GitHub](#)

- Built an autonomous AI agent using ReAct architecture capable of reasoning, tool usage, and web information retrieval.
- Implemented multi-step task execution using LangGraph workflows.
- Integrated LLM APIs, external search tools, and FastAPI services.
- Developed interactive UI for real-time conversational research support.
- Stack : Python, LangGraph, FastAPI, Streamlit, OpenAI, Groq, Tavily

[AuditShakti : AI-Powered Audit Intelligence Platform](#) | [GitHub](#)

- Designed an AI-driven compliance and audit assistant to automate document review, risk assessment, and audit report generation. Developed proof-of-concept workflows to improve efficiency in regulatory and compliance analysis.
- Stack : Python, LangChain, Streamlit, Vector Databases, LLMs

Skills

- **Programming** : Python, SQL, C++
- **Backend & APIs** : FastAPI, Flask, REST APIs, ETL Pipelines, Logging & Monitoring
- **Databases** : PostgreSQL, MySQL, Elasticsearch, Pinecone
- **AI & ML** : Machine Learning, NLP, LLMs, RAG, LangChain, LangGraph, Hugging Face, Ollama, OpenAI APIs
- **Healthcare Tech** : QT, VTK, ITK, Medical Imaging, Clinical Software Systems
- **Dev Tools** : Git, GitHub, Linux, Docker, Azure DevOps, PgAdmin, Power BI

Publications

1. Mental Health Risk Detection via Explainable AI (XAI): A Multi-Model Benchmark with Contextual Embeddings

International Conference on Artificial Intelligence Trends and Pattern Recognition (ICAITPR 2026) - Manuscript Under Review

2. AI-Based Analysis of Hospital Bed Utilization Patterns for Patient Readmission Prediction

International Conference on Smart AI and Health Innovation Through Tamil for Holistic Advancement (SAHITHYA 2025), NITTTR Chennai (ISBN Conference Proceedings) - Proceedings (Accepted)

3. Design of PON for Hospital Building Management: Enhancing Data Transmission Using Free Space Optics (FSO)

International Conference on Human-Centric Smart Computing (ICHCSC 2025), Springer Nature - Proceedings (Accepted)

Education

Bachelor of Engineering (ECE) – Honours in Artificial Intelligence

11/2021 – 05/2025 | Tamil Nadu, India

IFET College of Engineering, Anna University

CGPA : **8.71** | First Class with Distinction | **Rank Holder** | Associate Secretary

Achievements & Awards

- Awarded the **Rank Holder Award** and **Academic Excellence Award** for outstanding academic performance.
- Recognized as the **Best Student** for consistent academic excellence, leadership contributions, and overall achievements.
- Received the **Best Innovation Project Award** at Project Expo'25 for developing an impactful technology-based solution.
- Awarded **Best Presentation** in the Technical Talk Competition for excellence in technical and research presentation.
- Secured **Runner-Up** position in the CEMP Project Expo 2022 for innovative project development and implementation.
- **Finalist** in a National-Level Mathematics Competition, demonstrating strong analytical and problem-solving abilities.
- Earned the LinkedIn **Top Voice Badge** on three occasions for sharing technical knowledge and industry insights.
- **Recognized** by C-DAC for outstanding work performance and contributions to software development projects.

Courses & Certifications

- **Data Analytics** Professional Certificate | Google
- **Power BI** Analyst | Microsoft
- **Python** for Data Science | IBM
- **Generative AI** (LinkedIn Learning) | Microsoft
- **Prompt Engineering** (Cognitive Class) | IBM
- **Software Engineering** Architecture | Accenture